

Section A:

1. What are the two reasons for using layered protocol?

- a) Layering of protocols provides well-defined interfaces between the layers, so that a change in one layer does not affect an adjacent layer.
- b) The protocols of a network are extremely complicated and designing them in layers makes their implementation more feasible.

2. What are the three criteria necessary for an effective and efficient network?

- Performance
- Reliability
- Security.

3. Which of the OSI layers handles each of the following?

a) Dividing the transmitted bit stream into frames.

Ans: Data Link Layer is responsible in OSI layers for dividing the transmitted bit stream into frames.

b) Determining which route through the subnet to use.

Ans: Network layer is responsible in OSI layers for Determining which route through the subnet to use.

4. For n devices in a network, what is the number of cable links required for a mesh, ring, bus, and star topology?

Mesh: cable links required for a mesh topology is 6.

Bus: cable links required for a bus topology is 3.

Star: cable links required for a star topology is 4.

Ring: cable links required for a ring topology is 4.

5. list out two differences and two similarities between OSI and TCP/IP?

a) Similarities:

- Both models divide the networking concepts into layers.
- Both models simplify the troubleshooting process by dividing complex functions into simpler components.

b) Differences:

- OSI has 7 layers, whereas TCP/IP has 4 layers.
- OSI model, the transport layer, is only connection-oriented, whereas the TCP/IP model is both connection-oriented and connectionless.

Section B:

1. List the framing methods, explain any two of them.

- a) **Character count:** Uses a field in the header to specify the number of characters that are present frame.
- b) **Flag bytes with byte stuffing:** We can say that character stuffing is addition of 1 additional byte if there is presence of ESC or flag in text.
- c) **Starting and ending flags, with bit stuffing.**
- d) **Physical layer coding violations.**

2. How the data link layer deals with errors, list the methods with example of their usage?

a) Error Correction.

- To include enough redundant information along with each block to enable the receiver to deduce what was the transmitted data.
- Uses error correcting codes.
- Used when error rate is high.
Ex. Wireless channels.

b) Error Detection.

- To include just enough redundancy to allow the receiver to deduce that an error has occurred and ask for retransmission.
- Uses error detecting codes.
- Used in highly reliable channels.
Ex. Fiber optic.

3. Differentiate between LAN and WAN.

- **LAN:** abbreviated from Local Area Network, is a network that covers a small geographical area such as homes, offices, and groups of buildings.
- **WAN:** abbreviated from Wide Area Network, is a network that covers larger geographical areas that can span the globe

4. What are the types of data flow? Differentiate between them?

Simplex: sender can send the data, but that sender cannot receive data.

Half duplex: can be sender or receiver, but not at the same time.

Full duplex: sender and receiver at the same time.

5. The following character encoding is used in a data link protocol:

A: 01000111; B:11100011; FLAG: 01111110; ESC: 11100000 Show the bit Sequence transmitted (in binary) for the four-character frame: A B ESC FLAG when Flag bytes with byte stuffing framing method is used.

01111110 01000111 11100011 11100000
11100000 11100000 01111110 01111110